# Technical Reproducibility Validation Report

# STRsearch

Version c70179b
Autosomal STR

| | |
|---|---|
| Verification date | 2026-11-06 |
| Repository commit | c70179b3b175adc82a7314409af06900b3861d61 |
| Result | **Not verified as documented** |
| Reached | Runs + Plausible output |
| Permalink | https://strhub.app/verified/strsearch |
| STRhub | https://strhub.app |

## SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

### WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

### NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

## 1. Executive Summary

| | |
|---|---|
| **PURPOSE** | Verify that the tool installs, runs end-to-end, and produces structurally valid output. |
| **RESULT** | **Not verified as documented** |
| **REACHED** | Runs + Plausible output, 6/6 gates passed |
| **WHY** | The tool installed and its run produced its documented output. |
| **SCOPE** | Installation, execution, and output structure verification. |
| **NOT EVALUATED** | Genotype accuracy · Concordance · Forensic validity · Regulatory compliance |

## 2. What STRhub had to do to run this tool

STRhub wrote its own recipe for this tool, an environment and a command of its own rather than the repository's instructions, and ran that. Each item below is something a first-time user following the README would have to work out for themselves, and so a recommendation to the author. This run does not change the tool's label: the label is what happens as it is in the repository.

– Builds the environment on ubuntu 22.04 at the version floors the README declares (bwa, samtools, bamToFastq, seqtk). Instead of: The repository's own Dockerfile: ubuntu 16.04 with exact pins and Miniconda 4.3.31. The pinned stack no longer resolves.

– Fills in conf.py with the paths of the installed tools. Instead of: conf.py as shipped, with 'YOUR PATH/...' placeholders the user is to replace.

– Leaves usearch out. Instead of: usearch, which conf.py names. Licensed and unobtainable; only --assemble_pairs reaches it, and this run does not use it.

– Supplies an hg38 regions configuration in STRsearch's eleven-column layout, with flanking sequences, written by STRhub. Instead of: The repository's example, which is hg19 and covers five markers. No hg38 configuration ships with the repository.

With those changes, the run reached: **Runs + Plausible output**.

## 3. Verification Gates

| | | |
|---|---|---|
| **Available** | The pinned public source exists | PASS |
| **Installs** | The environment builds from source | PASS |
| **Starts** | The installed program answers --help in the built environment | PASS |
| **Runs** | Executes end-to-end without crashing | PASS |
| **Expected IO** | Produces a non-empty file in the declared format | PASS |
| **Plausible Output** | Output structure matches expected genotype-bearing format | PASS |
| Execution log | strsearch.log-external.txt | |

## 4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRsearch c70179b · commit c70179b
$ python3 /opt/STRsearch/pipeline.py \
--type paired \
--num_processors 4 \
--reads_threshold 30 \
--stutter_ratio 0.5 \
from_bam \
--working_path /data/out \
--sample sample \
--sex female \
--bam /data/in/input.bam \
--ref_bed /data/in/regions.bed \
--genotypes /data/out/genotypes.txt \
--multiple_alleles /data/out/multiple_alleles.txt \
--qc_matrix /data/out/qc_matrix.txt
```

## 5. Reproducibility Metadata

| | |
|---|---|
| Tool | STRsearch |
| Version | c70179b |
| Repository | https://github.com/AnJingwd/STRsearch |
| Commit (pinned) | c70179b3b175adc82a7314409af06900b3861d61 |
| Container OS | ubuntu-22.04 |
| Marker panel | Autosomal STR |
| Verified on | 2026-11-06 16:40:37 UTC |
| CI run | https://github.com/Tfronta/strhub-verified/actions/runs/42254219171 |

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

## 6. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

## 7. Output Content Evidence

| | |
|---|---|
| Output file | multiple_alleles.txt |
| Format | TSV |
| Sequence records | 388 |
| Distinct STR loci | 24 |
| Total reads | 2,401 |
| Max depth (single locus) | 51 |

## 8. Verification Data

The repository ships example data (35 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

| | |
|---|---|
| Dataset | Illumina BAM (hg38), NA12878 (autosomal, female) |
| Source | https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat… |
| License | Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider. |
| Ref. genome | GRCh38 / hg38 |
| Loci tested | CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA |
| Regions BED | Provided by STRhub |

## 9. Verification Matrix

| | | |
|---|---|---|
| N/A | Run on repository test data | — |
| | Run on reference dataset | Illumina BAM (hg38), NA12878 (autosomal, female) |
| | README check (5/5) | Advisory: all items present |
| | Install / environment setup | Present |
| | Run command | Present |
| | Expected input format | Present |
| | Produced output format | Present |
| | Dependencies / versions | Present |

## 10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, supplied with the submission.

## 11. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

## 12. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

## 13. Conclusion

- **Runs end-to-end**

  STRsearch c70179b installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

- **Produces plausible output**

  The tool generated a structurally valid TSV output file with genotype calls across 24 target forensic STR loci, with a maximum read depth of 51 at vWA.

- **Verdict: Runs**

  The tool installed and its run produced its documented output.

- **Repository test data not used**

  The repository ships example data (35 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

## Appendix: Detected Markers with Read Depth

STRsearch c70179b · verified 2026-11-06

| LOCUS | READ DEPTH | | LOCUS | READ DEPTH | |
|---|---|---|---|---|---|
| **vWA** | 241 | ■■■■■■■■■■■■■■■■■ | **CSF1PO** | 92 | ■■■■■■ |
| **D21S11** | 187 | ■■■■■■■■■■■■ | **D6S1043** | 92 | ■■■■■■ |
| **FGA** | 164 | ■■■■■■■■■■ | **D2S441** | 88 | ■■■■■ |
| **D19S433** | 131 | ■■■■■■■■ | **D1S1656** | 81 | ■■■■■ |
| **SE33** | 125 | ■■■■■■■■ | **PentaE** | 78 | ■■■■■ |
| **D18S51** | 123 | ■■■■■■■■ | **D13S317** | 66 | ■■■■ |
| **D16S539** | 105 | ■■■■■■ | **D10S1248** | 65 | ■■■■ |
| **D3S1358** | 99 | ■■■■■■ | **D2S1338** | 62 | ■■■■ |
| **PentaD** | 99 | ■■■■■■ | **D8S1179** | 62 | ■■■■ |
| **D7S820** | 96 | ■■■■■■ | **TPOX** | 55 | ■■■ |
| **D22S1045** | 95 | ■■■■■■ | **D12S391** | 55 | ■■■ |
| **D5S818** | 94 | ■■■■■■ | **TH01** | 46 | ■■■ |